

Q) W A C++ code which include a class student

find out the space occupied by object

→ C++ code to find space occupied by object

```
#include <iostream>
```

```
using namespace std;
```

```
class Student
```

```
{  
    int roll;
```

```
    char name[30];
```

```
    float marks;
```

```
public:  
    void getData()
```

```
{
```

```
    cout << "Roll";
```

```
    cin >> roll;
```

```
    cout << "name";
```

```
    cin >> name;
```

```

cout << "Marks";
cin >> marks;
}
void DisplayData()
{
    cout << "Roll : " << roll << endl;
    cout << "Name : " << name << endl;
    cout << "Marks : " << marks << endl;
}
void showSize()
{
    cout << "space = " << sizeof(*this);
};
};

int main()
{
    Student s;
    s.getData();
    s.DisplayData();
    s.showSize();
    return 0;
}

```

O/p : Roll : 101
 name : Rajasri
 Marks : 95.5
~~Name~~
 Roll : 101
 Name : Rajasri
 Marks : 95.5
 Space : 40 bytes

⇒ roll → 4 bytes , name → 30 bytes , marks → 4 bytes

↓
padding → 2 bytes

Starts a memory address that is efficient for CPU

⇒ this → points current object

*this → means current object itself

⇒ object name also can be written but object

name is not known so *this is used

⇒ for an empty class — size, byte